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# SDN Quiz

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**Result of SDN Quiz:**

| Total Questions | Full Score | Passing Rate | Your Score | Correct Answer Percentage | Elapsed |
|---|---|---|---|---|---|
| 8 | 110 | 80% | 110 | 100% | 00:02:22 |

Congratulations!

**You passed this test!**

If you want to retake this quiz, please press Ctrl + F5 on Windows or press CMD + R on Mac.

Your answers are shown below:

## Question 1

What are two benefits of controller-based networking compared to traditional networking? (Choose two)

- A. controller-based increases network bandwidth usage, while traditional lightens the load on the network.
- **B. controller-based reduces network configuration complexity, while traditional increases the potential for errors**
- **C. controller-based provides centralization of key IT functions, while traditional requires distributes management function**
- D. controller-based allows for fewer network failure, while traditional increases failure rates
- E. controller-based inflates software costs, while traditional decreases individual licensing costs

## Question 2

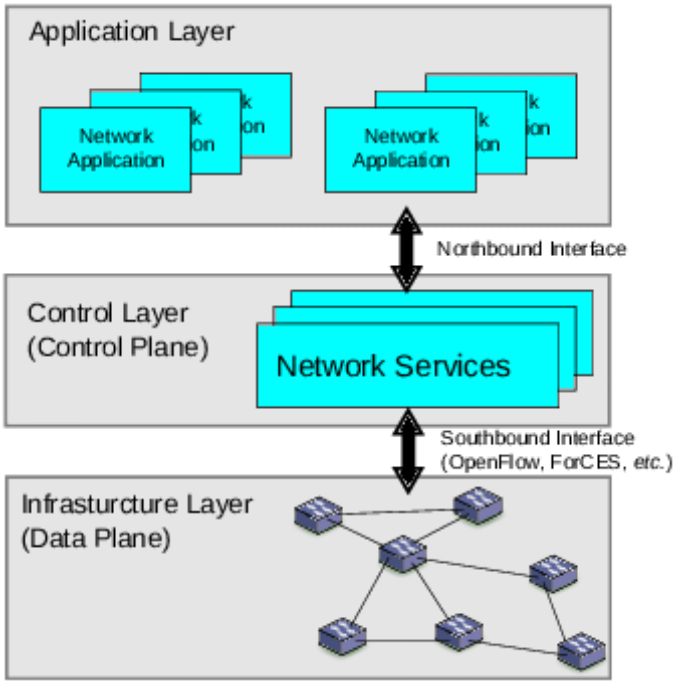
In software defined architectures, which plane is distributed and responsible for traffic forwarding?

- A. management plane
- B. control plane
- C. policy plane
- **D. data plane**

## Explanation

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The O n Networking Foundation identifies three main parts of the Software-defined networking (SDN Appli ion layer; Control layer and Infrastructure layer. SDN separates a router's control plane from th a rding) plane. The control plane makes routing decisions. The data plane forwards data (packets) thr the router. With SDN routing, decisions are made remotely instead of on each individual router.




## Question 3

What software defined architecture plane assists network devices with making packet forwarding decisions by providing Layer 2 reachability and Layer 3 routing information?

- A. policy plane
- B. management plane
- C. data plane
- **D. control plane**

## Explanation

**Control Plane Function**
In its simplest form, the control plane provides layer-2 MAC reachability and layer-3 routing information to network devices that require this information to make packet forwarding decisions. In the case of firewalls, the control plane would include stateful flow information for inspection. Control plane functionality can implemented as follows:

+ Distributed – Conventional routers and switches operate using distributed protocols for control, i.e. where each device makes its own decisions about what to do, and communicate relevant information to other devices for input into their decision making process. For example, the Spanning Tree Protocol (STP), Fabric Path, and routing protocols such as IS-IS and BGP provide distributed control of packet forwarding functionality to networking devices.

+ Centralized – In this case, a centralized controller provides the necessary information for a network element to make a decision. For example, these controller(s) instruct networking devices on where to forward packets by explicitly programming their MAC and FIBs.

Reference: https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data_Center/VMDC/SDN/SDN.html

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## ...tion 4

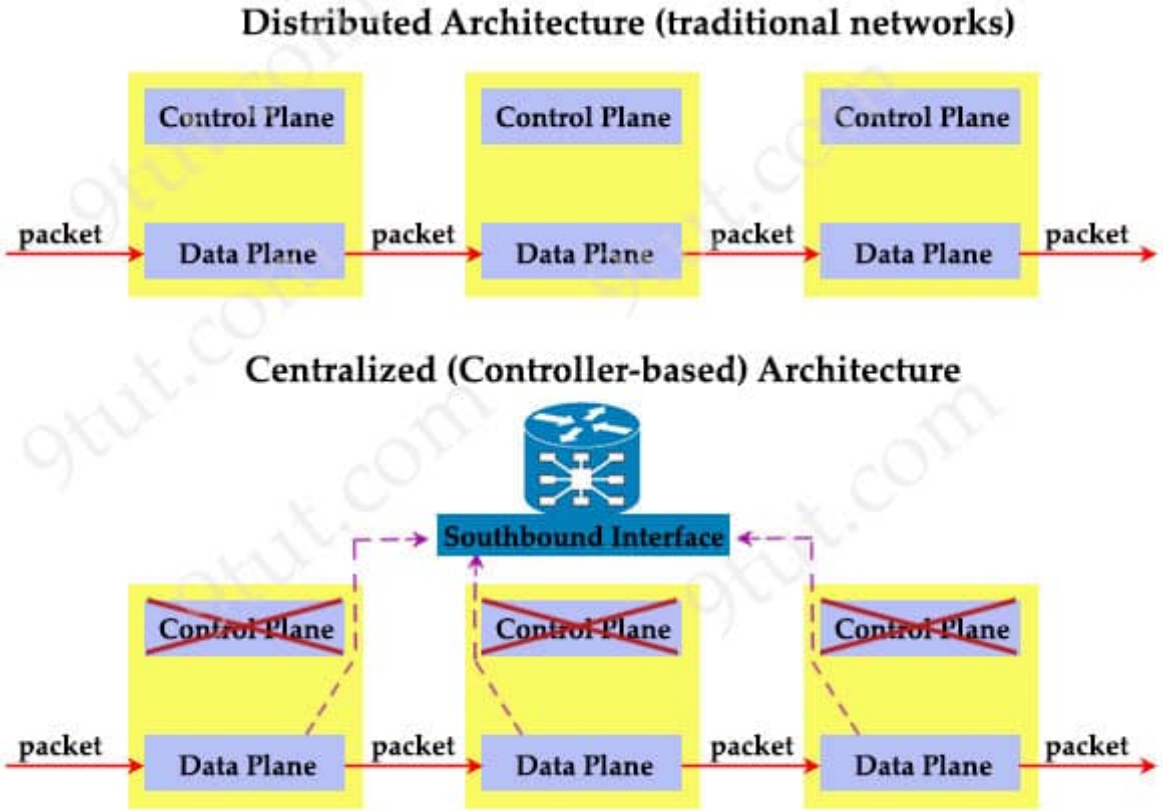
Which statement correctly compares traditional networks and controller-based networks?

- A. Traditional and controller-based networks abstract policies from device configurations
- B. Only traditional networks offer a centralized control plane
- C. Only traditional networks natively support centralized management
- **D. Only controller-based networks decouple the control plane and the data plane**

### Explanation

Most traditional devices use a distributed architecture, in which each control plane is resided in a networking device. Therefore they need to communicate with each other via messages to work correctly.

In contrast to distributed architecture, centralized (or controller-based) architectures centralizes the control of networking devices into one device, called **SDN controller** -> Answer 'Only controller-based networks decouple the control plane and the data plane' is correct.




## Question 5

Which statement about the Cisco ACI fabric is most accurate?

- **A. An APIC is a cluster of at least three APIC controllers, providing a single point of management without a single point of failure.**
- B. The fabric header carries the EPG from the egress to the ingress leaf switch.
- C. The APIC is able to enforce security by inserting itself into the data path.
- D. The spine switch rewrites the EPG from ingress to egress when it performs the forwarding proxy function

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## tion 6

What are two southbound APIs? (Choose two)

- **A. Open Flow**
- **B. NETCONF**
- C. Thrift
- D. DSC
- E. CORBA

### Explanation

OpenFlow is a well-known southbound API. OpenFlow defines the way the SDN Controller should interact with the forwarding plane to make adjustments to the network, so it can better adapt to changing business requirements.

The Network Configuration Protocol (NetConf) uses Extensible Markup Language (XML) to install, manipulate and delete configuration to network devices.

Other southbound APIs are:
+ onePK: a Cisco proprietary SBI to inspect or modify the network element configuration without hardware upgrades.
+ OpFlex: an open-standard, distributed control system. It send "summary policy" to network elements.

## Question 7

What are two characteristics of a controller-based network? (Choose two)

- **A. It uses northbound and southbound APIs to communicate between architectural layers**
- B. It decentralizes the control plane, which allows each device to make its own forwarding decisions
- C. The administrator can make configuration updates from the CLI
- **D. It moves the control plane to a central point**
- E. It uses Telnet to report system issues.

## Question 8

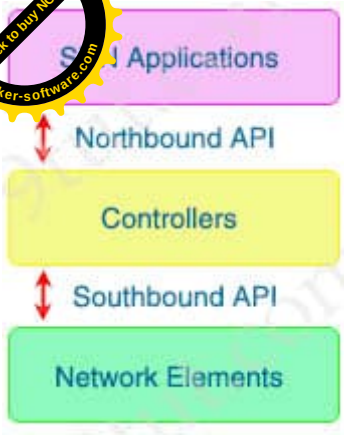
Which API is used in controller-based architectures to interact with edge devices?

- A. underlay
- **B. southbound**
- C. northbound
- D. overlay

### Explanation

The Southbound API is used to communicate with network devices.

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Comments (3) Comments

1. Nabeel
   December 24th, 2020

   I see the tutorial is very brief. Is it because the exam only covers SDN very briefly or is there a more detailed tutorial you'd recommend ?

2. mano
   November 25th, 2021

   you should make it more complicated it seems very easy for the ccna exam prep

3. Jacob B
   April 19th, 2022

   Done

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- === New CCNA v1.1 Lab Sims (very important) ===
- Voice VLAN Configuration Sim 2
- Voice VLAN & LLDP Sim
- Allowed, Native VLAN & LACP Sim
- Static Routing Configuration Sim 4
- IPv4 and IPv6 Connectivity Sim
- VLAN & CDP Sim 2
- IPv4 and IPv6 Assignment Sim
- VLAN CDP & LLDP Sim
- OSPF Configuration Sim 2
- OSPF Configuration Sim 3
- Trunking & LACP Sim
- VLAN Configuration Sim
- Allowed, Native VLAN & LACP Sim 2
- 802.1Q Trunking & LACP
- IPv4 and IPv6 Assignment Sim 2
- 802.1Q Trunking, Native VLAN & LACP
- Static Routing Configuration Sim 5
- Static Routing Configuration Sim 6
- PAT, NTP & DHCP Relay Sim
- OSPF DR Sim
- NACL DHCP Snooping 3
- NAT DHCP SSH Sim
- ==== Lab Sims for CCNA (although they are rather old but please learn all of them too) ====
- IP Services Sim Version 2
- OSPF Configuration Sim
- LACP Configuration Sim
- Voice VLAN Configuration Sim
- VLAN and Trunking Configuration Sim
- Named Access-list & Port Security Sim
- Named Access-list & DHCP Snooping Sim
- VLAN and CDP Sim
- IPv4 and IPv6 Deployment Sim
- Static Routing Configuration Sim 3
- VLAN & LLDP Sim
- IPv6 Static Routing Sim
- Named Access-list & DHCP Snooping Sim 2
- ISL Trunking and LACP Sim
- === Below Sims are old and retired so you do not need to learn them ===
- Static Routing Configuration Sim 2
- IP Services Sim
- Static Routing Configuration Sim

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- Routing Questions
- Routing Questions 2
- OSPF Questions
- OSPF Questions 2
- EIGRP Questions
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- Syslog Questions
- HSRP Questions
- Access-list Questions
- AAA Questions
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